3. MAGNETIC FIELD ALONG THE AXIS OF A CIRCULAR COIL-DETERMINATION OF B_H

AIM

To determine the horizontal component of Earth's magnetic field using current carrying circular coil and deflection magnetometer.

APPARATUS REQUIRED

Circular coil apparatus, compass box, rheostat, battery or power supply, ammeter, commutator, key and connecting wires.

FORMULA

$$B_H = \frac{\mu_0 n I r^2}{2\left(r^2 + x^2\right)^{3/2}} \left(\frac{1}{\tan\theta}\right) \text{ (Tesla)}$$

where, B_H → Horizontal component of Earth's magnetic field (T)

μ_o → Permeability of the free space ($4\pi \times 10^{-7}$ H m^{-1})

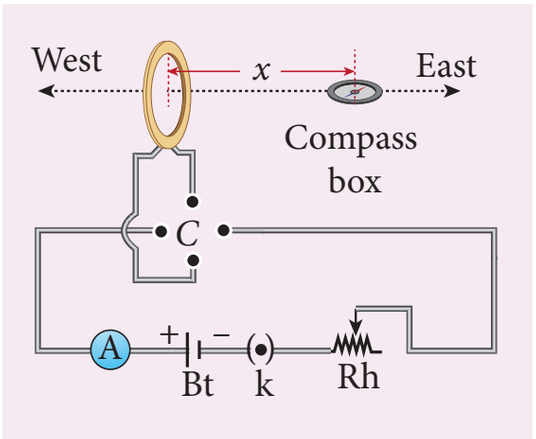
n → Number of turns included in the circuit (No unit)

I → Current flowing through the coil (A)

r → Radius of the circular coil (m)

x → Distance between center of compass box and centre of the coil (m)

CIRCUIT DIAGRAM



PROCEDURE

- The preliminary adjustments are carried out as follows.
  - i. The leveling screws are adjusted so that the circular coil is vertical.
  - ii. The wooden bench is adjusted to be along the magnetic east-west direction i.e., along aluminium pointer.
  - iii. The circular coil is rotated so that its plane is in magnetic meridian i.e., along the north-south direction.
  - iv. A compass box is placed with its centre coinciding with the axis of the coil.
  - v. The compass box alone is rotated till the aluminium pointer reads 0° – 0°
- Electrical connections are made as shown in the circuit diagram.
- The compass box is placed along its axis, with its centre at a distance x from the centre of the coil on one side.

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